

Qatar University

Course Master Syllabus التوصيف الرئيسي للمقرر

STAT 481 — Multivariate Analysis — Fall 2026

Section L01 (CRN 16064) — Sun/Tue/Thu 12:00–12:50

Course Number رمز المقرر	STAT 481	Course Title (Arabic) اسم المقرر باللغة العربية	التحليل المتعدد
		Course Title (English) اسم المقرر باللغة الانجليزية	Multivariate Analysis
Course Designation نوع المقرر	LC: Lecture	Course Term Offering الفصل الدراسي لطرح المقرر	☒ Fall ☐ Spring
Number of Credit Hours عدد الساعات المكتسبة	3	Number of Contact Hours عدد الساعات التدريسية	LC: Lecture 3 LB: Laboratory —
Course Package (English) المجموعة الدراسية باللغة الإنجليزية	Major required course for B.Sc. in Statistics	New course (not existed in Banner)? مقرر جديد (غير موجود في بانر)؟	Yes ☐ No ☒
Course Package (Arabic) المجموعة الدراسية باللغة العربية	مقرر المتطلبات الأساسية لدرجة البكالوريوس في الإحصاء	Language of Instruction لغة التدريس	Arabic ☐ English ☒ Both ☐
Required إجباري	Yes ☒ No ☐	Elective اختياري	Yes ☐ No ☒
Course Prerequisites متطلب سابق	STAT 322 and MATH 231	Course Co-requisites متطلب متزامن	None
Catalog Description وصف المقرر	This course is a compulsory course in statistics. It covers the multivariate topics: Organization of Multivariate Data; Multivariate Distributions; Mahalanobis Distance; Hotelling's T^2; Multivariate Analysis of Variance and Regression; Data Reduction Techniques; Discriminant and Classification Analysis; and Canonical Correlation Analysis. Statistical software is used throughout for the analysis of real multivariate data sets. The course is delivered using student-centered learning and digitally enriched approaches, and it is research-informed.		
Textbook(s) الكتب الدراسية	Required Textbook: Applied Multivariate Statistical Analysis, Richard A. Johnson and Dean W. Wichern, 6th edition, Pearson / Prentice Hall (main textbook). Suggested Additional Resources: • An Introduction to Applied Multivariate Analysis with R, B. Everitt and T. Hothorn, 2011, Springer (R companion for the course). • Methods of Multivariate Analysis, A. C. Rencher, 2nd Edition, 2002, John Wiley and Sons. • Applied Multivariate Statistical Analysis, W. K. Härdle and L. Simar, 3rd edition, 2012, Springer. • Multivariate Statistical Methods: A Primer, B. Manly, 2nd edition, 1994, Chapman & Hall/CRC. • An Introduction to Multivariate Statistical Analysis, T. W. Anderson, 3rd Edition,		

	2003, Wiley-Interscience. Applied Multivariate Methods for Data Analysis, Dallas E. Johnson, 1998, Duxbury Press. Updated lecture notes (L1–L26), practice sheets, sample quizzes, and R/Quarto scripts are posted on Blackboard.
Course Objectives أهداف المقرر	Obj. 1: Enabling the students to organize multivariate data into an array and calculate its mean vector, covariance matrix, and generalized variance. Obj. 2: Introducing the students to the Multivariate Normal (MN) distribution and the distribution of the sample mean and covariance from an MN distribution and the associated inferences. Obj. 3: Helping students acquire knowledge of principal components analysis (PCA) and factor analysis (FA). Obj. 4: Introducing the elements of discriminant analysis and canonical correlation. Obj. 5: Familiarizing the students with cluster analysis and multidimensional scaling concepts, and training the students to use statistical packages to analyze multivariate data sets.
Course Learning Outcomes مخرجات التعلم للمقرر	- CLO 1: Be familiar and comfortable with matrix notations and terminology; be able to perform elementary matrix operations and understand more advanced matrix operations. - CLO 2: Be able to summarize a set of data with multiple variables using indices of bivariate relation or association, including covariance and correlation matrices. - CLO 3: Calculate statistical quantities like the multivariate mean, covariance matrix and the generalized variance. - CLO 4: Test hypotheses about the parameters of the multivariate normal distribution and compare several multivariate normal means; perform principal component analyses and factor analyses. - CLO 5: Apply the techniques of canonical correlation and discriminant analysis, also cluster analysis and multidimensional scaling; use statistical packages to analyze multivariate data.

WEEKLY TOPICS — المخطط الأسبوعي (FALL 2026: SUN/TUE/THU, AUG 23 - DEC 3)

Semester structure: 14 teaching weeks (42 sessions). Each block of lectures ends with an in-class quiz; every quiz is preceded by one or two practice sessions in which we solve a practice sheet and correct a sample quiz, so students know exactly the level and format expected. The midterm is preceded by two revision sessions in the same spirit. R laboratory sessions are integrated into the lecture sequence. Mid-Fall Break: Oct 25–29 (no classes).

Week	Date (2026)	Day	Topic / Session	Reading / Materials	Notes / Assessment
1	Aug 23	Sun	Course introduction: syllabus, assessment plan, course policies; how lectures, R labs, practice sessions and quizzes are organized	Syllabus	—
1	Aug 25	Tue	Lecture 1: Aspects of multivariate analysis — the multivariate data array, applications and organization of data	J&W Ch. 1; Notes L1	—
1	Aug 27	Thu	Lecture 2: Summary statistics — mean vector, covariance and correlation matrices, generalized variance	J&W §1.3–1.4; Notes L2	End of Add/Drop
2	Aug 30	Sun	Lecture 3 (R lab): R, RStudio and Quarto; importing data, matrix operations and summary statistics in R	Notes L3	R Lab 1 out

Week	Date (2026)	Day	Topic / Session	Reading / Materials	Notes / Assessment
2	Sep 1	Tue	Lecture 4: Graphical displays — scatterplot matrices, three-dimensional plots, stars and Chernoff faces	J&W §1.4; Notes L4	—
2	Sep 3	Thu	Lecture 5: Distance — Euclidean distance and statistical (Mahalanobis) distance	J&W §1.5; Notes L5	—
3	Sep 6	Sun	Practice session: Practice Sheet 1 (organization of data, summary statistics, distance)	Practice Sheet 1	R Lab 1 due
3	Sep 8	Tue	Practice session: correction of Sample Quiz 1	Sample Quiz 1	Quiz preparation
3	Sep 10	Thu	QUIZ 1 (in class): organization of multivariate data, summary statistics & distance (Weeks 1–2)	—	In-class quiz
4	Sep 13	Sun	Lecture 6: Matrix algebra — transpose, products, inverse, determinant, trace and rank	J&W §2.2–2.3; Notes L6	—
4	Sep 15	Tue	Lecture 7: Positive definite matrices and quadratic forms; spectral decomposition, eigenvalues and eigenvectors	J&W §2.3; Notes L7	—
4	Sep 17	Thu	Lecture 8 (R lab): matrix algebra, eigen-decomposition and generalized variance in R	Notes L8	R Lab 2 out
5	Sep 20	Sun	Lecture 9: Random vectors — mean vectors, covariance matrices, partitioning, linear combinations of variables	J&W §2.5–2.6; Notes L9	—
5	Sep 22	Tue	Lecture 10: Sample geometry and random sampling; expected values of the sample mean and covariance matrix	J&W §3.1–3.4; Notes L10	—
5	Sep 24	Thu	Practice session: Practice Sheet 2 (matrix algebra, random vectors and sample geometry)	Practice Sheet 2	R Lab 2 due
6	Sep 27	Sun	Practice session: correction of Sample Quiz 2	Sample Quiz 2	Quiz preparation
6	Sep 29	Tue	QUIZ 2 (in class): matrix algebra, random vectors & sample geometry	—	In-class quiz
6	Oct 1	Thu	Lecture 11: The multivariate normal distribution — density, properties and the bivariate normal	J&W §4.2; Notes L11	—
7	Oct 4	Sun	Lecture 12: Marginal and conditional distributions; sampling distribution of $\bar{X}$ and S ; the Wishart distribution	J&W §4.3–4.4; Notes L12	—
7	Oct 6	Tue	Lecture 13 (R lab): assessing normality — Q-Q plots, chi-square plots and transformations to near normality	J&W §4.6–4.8; Notes L13	R Lab 3 out
7	Oct 8	Thu	Practice session: Practice Sheet 3 (multivariate normal distribution and	Practice Sheet 3	—

Week	Date (2026)	Day	Topic / Session	Reading / Materials	Notes / Assessment
			assessment of normality)		
8	Oct 11	Sun	Lecture 14: Inferences about a mean vector — Hotelling's T^2 and the likelihood ratio test	J&W §5.2–5.3; Notes L14	—
8	Oct 13	Tue	Lecture 15: Confidence regions, simultaneous confidence intervals and Bonferroni intervals; large-sample inference	J&W §5.4–5.6; Notes L15	R Lab 3 due
8	Oct 15	Thu	Lecture 16 (R lab): Hotelling's T^2 , confidence regions and simultaneous intervals in R	Notes L16	Last day to withdraw from a course: Thu, Oct 15
9	Oct 18	Sun	Midterm revision I: structured problem solving (Weeks 1–8)	Revision sheet	Midterm preparation
9	Oct 20	Tue	Midterm revision II: correction of a sample midterm	Sample midterm	Midterm preparation
9	Oct 22	Thu	MIDTERM EXAM (in class): Weeks 1–8	—	Midterm (30%)
—	Oct 25	Sun	No class — Mid-Fall Break	—	Break
—	Oct 27	Tue	No class — Mid-Fall Break	—	Break
—	Oct 29	Thu	No class — Mid-Fall Break	—	Break
10	Nov 1	Sun	Lecture 17: Comparison of several multivariate means — paired comparisons and repeated measures designs	J&W §6.2–6.3; Notes L17	Project topics assigned
10	Nov 3	Tue	Lecture 18: Comparing two population mean vectors; the one-way MANOVA model	J&W §6.3–6.4; Notes L18	—
10	Nov 5	Thu	Lecture 19: MANOVA test statistics (Wilks' Λ , Pillai, Lawley–Hotelling, Roy); assumptions and diagnostics; two-way MANOVA	J&W §6.4–6.6; Notes L19	Last day to withdraw from the semester: Thu, Nov 5
11	Nov 8	Sun	Lecture 20 (R lab): one-way and two-way MANOVA in R; checking assumptions	Notes L20	R Lab 4 out
11	Nov 10	Tue	Practice session: Practice Sheet 4 (inference about mean vectors and MANOVA) + correction of Sample Quiz 3	Practice Sheet 4; Sample Quiz 3	Quiz preparation
11	Nov 12	Thu	QUIZ 3 (in class): inference about mean vectors & MANOVA	—	In-class quiz
12	Nov 15	Sun	Lecture 21: Principal component analysis — population and sample principal components, interpretation	J&W §8.2–8.3; Notes L21	R Lab 4 due
12	Nov 17	Tue	Lecture 22 (R lab): number of components, PCA on the correlation matrix, scree plots and biplots in R	J&W §8.3–8.5; Notes L22	R Lab 5 out
12	Nov 19	Thu	Lecture 23: Factor analysis — the orthogonal	J&W §9.2–9.3;	—

Week	Date (2026)	Day	Topic / Session	Reading / Materials	Notes / Assessment
			factor model; principal component and maximum likelihood estimation	Notes L23	
13	Nov 22	Sun	Lecture 24 (R lab): rotation methods, factor scores and a strategy for factor analysis; factor analysis in R	J&W §9.4–9.6; Notes L24	—
13	Nov 24	Tue	Practice session: Practice Sheet 5 (principal components and factor analysis) + correction of Sample Quiz 4	Practice Sheet 5; Sample Quiz 4	R Lab 5 due; quiz preparation
13	Nov 26	Thu	QUIZ 4 (in class): principal components & factor analysis	—	In-class quiz
14	Nov 29	Sun	Lecture 25: Canonical correlation analysis — canonical variates and correlations, interpretation, redundancy analysis	J&W Ch. 10; Notes L25	—
14	Dec 1	Tue	Lecture 26: Cluster analysis (hierarchical and K-means) and multidimensional scaling; elements of discrimination and classification	J&W §11.2–11.3, §12.3–12.6; Notes L26	Project report due
14	Dec 3	Thu	Project presentations; final revision and exam guidance	Review notes	Last day of classes
—	Dec 6–17	—	FINAL EXAM period (comprehensive; exact date, time and venue set by the Registrar)	—	Final exam (35%)

EDUCATION EXCELLENCE THEMES — أطر التميز في التعليم

*The themes “Digitally Enriched” and “Learner-Centric” are included in all courses; “Research-Informed” is the prioritized additional theme for this course.

Themes المحاور	Implemented تم تنفيذ المحور	Teaching Methods طرق التدريس	Course Learning Outcomes مخرجات التعلم للمقرر
Digitally Enriched*	Yes ☒ No ☐	• Live lecture recordings. • Blackboard assignments and quizzes. • Use of R, RStudio and Quarto. • R scripts and notes on Blackboard.	• CLO 1 • CLO 2 • CLO 3 • CLO 4 • CLO 5
Learner-Centric*	Yes ☒ No ☐	• Discussion: involving students in learning by initiating discussions. • Problem-solving sessions based on practice sheets and sample quizzes.	• CLO 1 • CLO 2 • CLO 3 • CLO 4 • CLO 5

Themes المحاور	Implemented تم تنفيذ المحور	Teaching Methods طرق التدريس	Course Learning Outcomes مخرجات التعلم للمقرر
		• Flipped-classroom sessions on selected topics. • Pair work on the course project.	
Experiential	Yes ☐ No ☒	—	—
Entrepreneurial	Yes ☐ No ☒	—	—
Research-Informed	Yes ☒ No ☐	• Real-data modelling examples from published research. • Reading and discussion of a research paper. • Semester project on a real multivariate data set.	• CLO 1 • CLO 2 • CLO 3 • CLO 4 • CLO 5

GRADUATE ATTRIBUTES — سمات الخريجين

Graduate Attributes	Supporting Competences	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5	CLO 6	CLO 7	CLO 8
A1: Competent	C1: Subject-matter mastery	X	X	X	X	X			
	C2: Critical-thinking skills	X			X				
	C3: Problem-solving skills			X		X			
	C4: Research, and Novel and Adaptive Thinking								
A2: Life-long Learner	C5: Self-awareness	X			X				
	C6: Adaptability		X	X		X			
	C7: Adaptive Thinking				X	X			
	C8: Desire for life-long learning								
A3: Well Rounded	C9: Cultured								
	C10: Effective communication skills			X		X			
	C11: Awareness of local and international issues								
A4: Ethically and Socially Responsible	C12: Embody the Arabic-Islamic identity								

Graduate Attributes	Supporting Competences	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5	CLO 6	CLO 7	CLO 8
	C13: Embrace diversity								
	C14: Professional and ethical conduct		X		X				
	C15: Civically engaged								
	C16: Community and Global Engagement			X					
A5: Entrepreneurial	C17: Creativity and innovation								
	C18: Collaborative			X					
	C19: Management								
	C20: Interpersonal			X					
	C21: Leadership								

FACULTY INFORMATION

Instructor	Prof. Dhafer Malouche
Office Location	BCR-D 203
Office Hours	Tuesdays 10:55–11:55, room BCR-D 203, or by appointment
Office Telephone	(+974) 4403 4628
E-mail	dhafer.malouche@qu.edu.qa

CLASSES INFORMATION

Section	CRN	Day(s) & Time(s)	Classroom	Campus
L01	16064	Sun/Tue/Thu 12:00–12:50 pm	BCR — Corridor H210	Male Designated Area

Semester Start Date: Sunday, August 23, 2026 (first day of classes).

Semester End Date: Thursday, December 3, 2026 (last day of classes). Final exams: December 6–17, 2026.

DELIVERY METHODS

The course will be delivered through a combination of active learning strategies. These will include:

- Slide lectures and active classroom-based participation.
- Hands-on practical experience through R labs and practice sessions.
- Collaborative learning through small-group problem solving and the course project.
- Flipped-classroom activities on selected topics prepared by students before class.
- Web-based learning through Blackboard (slides, practice sheets, sample quizzes, R scripts, grades).

COMPUTER SOFTWARE

No particular software licence is required. All computational work in this course is carried out in R (with RStudio); we will use packages such as MASS, car, psych, GGally, corrplot, factoextra, candisc and cluster to compute, simulate and visualize multivariate analyses. R laboratory reports and the course project are submitted as reproducible Quarto (.qmd) documents together with the rendered PDF or HTML output. All R scripts used in class are posted on Blackboard, and no prior programming experience is assumed.

ACADEMIC SUPPORT

The University Student Learning Support Center (SLSC) provides academic support services to male and female students at QU. The SLSC is a supportive environment where students can seek assistance with academic coursework, writing assignments, transitioning to college academic life, and other academic issues. SLSC programs include Peer Tutoring, the Writing Lab, Writing Workshops, and Academic Success Workshops. Students may also seek confidential academic counseling from the professional staff at the Center.

Contact: Tel (00974) 4403 3876 — Female Student Activities Building — learningcenter@qu.edu.qa

It is Qatar University policy to provide educational opportunities that ensure fair, appropriate and reasonable accommodation to students who have disabilities that may affect their ability to participate in course activities or meet course requirements. Students with disabilities are encouraged to contact their Instructor to ensure that their individual needs are met. The University, through its Special Needs section, will exert all efforts to accommodate individuals' needs.

Special Needs Section: Female (00974) 4403 3843, Male (00974) 4403 3854 — Student Activities Building — specialneeds@qu.edu.qa

COURSE REGULATIONS

Participation. Class participation and attendance are important elements of every student's learning experience at Qatar University, and the student is expected to attend all classes. A student should not miss more than 25% of the classes during a semester. Those exceeding this limit of 25% will receive a failing grade regardless of their performance. It is a student's responsibility to monitor the frequency of their own absences. Attendance record begins on the first day of class irrespective of the period allotted to drop/add and late registration. It is a student's responsibility to sign in; failure to do so will result in a non-attendance being recorded.

In exceptional cases, the student, with the instructor's prior permission, could be exempted from attending a class provided that the number of such occasions does not exceed the limit allowed by the University. The instructor will determine the acceptability of a reason for being absent. A student who misses more than 25% of classes and has a valid excuse for being absent will be allowed to withdraw from the course.

Calculators. Students are allowed to use a scientific calculator in quizzes and exams. Any calculator with power, root, logarithm and exponential functions is sufficient (Casio FX-82 ES Plus or equivalent). Graphing calculators are NOT permitted, and sharing calculators during quizzes and exams is not allowed.

Plagiarism. Plagiarism is considered a serious academic offence and can result in your work losing marks or being failed. QU expects its students to adopt and abide by the highest standards of conduct in their interaction with their professors, peers, and the wider University community. As such, a student is expected not to engage in behaviours that compromise his/her own integrity as well as that of Qatar University. Plagiarism includes the following examples, and it applies to all student assignments or submitted work:

- Use of the work, ideas, images or words of someone else without his/her permission or reference to them.
- Use of someone else's wording, name, phrase, sentence, paragraph or essay without using quotation marks.
- Misrepresentation of the sources that were used.

The instructor has the right to fail the coursework or deduct marks where plagiarism is detected.

Use of Artificial Intelligence (AI) tools. Quizzes, the midterm and the final exam are closed-book, in-class assessments: any use of AI tools or other unauthorized aids during these assessments is prohibited and will be treated as academic misconduct. For R laboratory reports and the course project, AI tools may be used only as a study aid (for example, to check reasoning or explore R code), and any such use must be acknowledged in the submitted report. The submitted work must be the student's own, written in the student's own words, and must show complete derivations, output and justifications — answers without visible work receive no credit. Students may be asked to explain any part of a submitted report in class or during office hours; inability to explain one's own submitted work will result in the loss of the corresponding marks.

Late or Missed Assessments. In all cases of assessment, students who fail to attend an exam or quiz on the scheduled date without prior permission, and/or are unable to provide a medical note, will automatically receive a fail grade for this part of the assessment. There is no makeup for the quizzes.

- Submitting laboratory reports and the project on time is a key part of the assessment process. Students who fail to submit their work by the deadline specified will automatically receive a 10% penalty. Work handed in more than 24 hours late will receive a further 10% penalty; each subsequent 24 hours will result in a further 10% penalty.
- In cases where a student misses an assessment on account of a medical reason or with prior permission, in line with University regulations an incomplete grade for the specific assessment will be awarded and an alternative assessment or extension can be arranged.

Student Complaints Policy. Students at Qatar University have the right to pursue complaints related to faculty, staff, and other students. The nature of the complaints may be either academic or non-academic. For more information about the policy and processes related to this policy, you may refer to the students' handbook.

COURSE ASSESSMENT

Evaluation Policy: The instructor may modify the evaluation policy and weights in case the University changes the teaching style from online teaching to regular teaching or vice versa, in a manner that does not harm the educational process or the interest of students.

Students will be graded through the following means of assessment, and their final grade will be calculated from the forms of assessment as listed below with their grade weighting taken into account.

Assessment	Grade Weighting	Remark
In-Class Quizzes*	15%	4 quizzes, taken in class on paper: Sep 10, Sep 29, Nov 12, Nov 26. Each quiz is preceded by one or two practice sessions (practice sheet + correction of a sample quiz).
Project and R Laboratory Work	15%	5 R laboratory reports submitted as reproducible Quarto documents: Lab 1 (Aug 30 → Sep 6), Lab 2 (Sep 17 → Sep 24), Lab 3 (Oct 6 → Oct 13), Lab 4 (Nov 8 → Nov 15), Lab 5 (Nov 17 → Nov 24); and a project, carried out in pairs, on a real multivariate data set (topics assigned Nov 1, report due Dec 1, presentations Dec 3).
Flipped-Classroom and In-Class Activities**	5%	Graded problems and discussions during lectures and practice sessions; selected topics prepared by students before class; some answers submitted via Blackboard.
Mid-Term Exam	30%	Thursday, October 22, 2026 (Week 9, in class), covering Weeks 1–8. Two revision sessions (structured problem solving + sample midterm correction) are scheduled on Oct 18 and Oct 20.
Final Exam	35%	Comprehensive; during the final exam period Dec 6–17, 2026 (exact date/venue announced by the Registrar).

* There will be 4 in-class quizzes. There is NO makeup for the quizzes.

** **Flipped-Classroom and In-Class Activities:** For selected topics, students study designated pages from the lecture notes and the textbook and consult related resources before coming to class; in class the instructor summarizes the topic, answers questions and poses problems for students to discuss and solve. Students solve specific problems/questions and sometimes will be asked to submit answers via Blackboard. These questions will be graded and included in the final grade.

Note: The course project is a required component of STAT 481. It is carried out in pairs on a real multivariate data set (subject to approval), applies at least two of the methods studied in the course, and is submitted as a reproducible Quarto report together with a short oral presentation.

Grades are not negotiable and are awarded according to the following criteria:

Letter Grade	Description	Percentage	Grade Points
A	Excellent	90–100	4.00
B+	Very Good	85–<90	3.50
B	Very Good	80–<85	3.00
C+	Good	75–<80	2.50
C	Good	70–<75	2.00
D+	Pass	65–<70	1.50
D	Pass	60–<65	1.00
F	Fail	Less than 60	0.00
I	Incomplete	—	—